

Titilayo Oshinowo

titiooshy@gmail.com | (518)-898-1238 | <https://www.linkedin.com/in/titilayo-oshinowo/>

EDUCATION

Franklin W. Olin College of Engineering

Bachelor of Engineering Computing and Design | GPA: 3.81

Needham, MA

Anticipated Graduation: May 2028

WORK EXPERIENCE

New York Power Authority

White Plains, NY

Digital End User Intern

June 2026 – Present

- Advanced NYPA's Digital Employee Experience (DEX) rollout by researching industry benchmarks and aligning initiatives with VISION2030 goals.
- Reduced manual IT touchpoints by supporting Microsoft Intune and Azure device provisioning and troubleshooting for enterprise end users.
- Informed digital platform pilot decisions by analyzing employee IT usage data in Excel and building trend-tracking dashboards.
- Improved cross-team alignment by coordinating DEX project updates between IT and business stakeholders using UX-driven communication.

PIECE Lab

Needham, MA

Research Assistant

May 2025 – Present

- Identified SEPTs (Software Engineering Practices & Tools) adoption gaps across STEM student projects by analyzing usage factors and their downstream design and maintenance effects.
- Developed a four-dimension framework (Context, Suitability, Understanding, Evaluation) to characterize student paper-reading strategies in engineering education.
- Surfaced software-knowledge gaps that informed improved SEPT instruction by conducting qualitative interview analysis in NVivo.
- Co-authored SIGCSE 2026 and FIE 2026 conference papers through faculty-led study design and dissemination.

Olin Assistive Technology Lab

Needham, MA

Device Interaction & Capabilities

January 2025 – Present

- Shaped core design decisions for a Braille and text-to-speech conversion tool by conducting user research and interviews with blind participants.
- Increased tool usability through iterative prototyping cycles driven directly by user feedback.
- Automated file-conversion workflows using Python, enabling scalable accessibility solutions in collaboration with designers and developers.

PROJECTS

The FitBot

Needham, MA

Embedded Systems & Wearable Health Device

October 2025 – December 2025

- Enabled real-time motion and vital-sign tracking on a resource-constrained wearable by developing Arduino firmware with closed-loop sensor calibration.
- Built a web-based dashboard for remote visualization and live trend analysis, integrating firmware, real-time processing, and UI/UX into one system.

Asteroid Family Identification using DBSCAN

Needham, MA

Data Science & Algorithm Design

February 2026 – March 2026

- Identified 7 Trojan asteroid families from 11,801 asteroids using a custom Zappalà-metric DBSCAN pipeline, benchmarked against the AstDys ground truth catalog.
- Achieved 100% completeness across all target families by tuning the DBSCAN cutoff via systematic parameter sweeps, outperforming hierarchical clustering baselines.

TCP Chat Application

Needham, MA

Systems Programming in C

February 2026 – March 2026

- Built a multithreaded C chat server using raw TCP sockets and POSIX threads to support real-time multi-client messaging.
- Ensured thread-safe broadcasting and clean shutdown via a mutex-protected client list and a graceful SIGINT handler.

Automatic Alt-Text Generation

Needham, MA

Vision-Language Model Fine-Tuning

April 2026 – May 2026

- Fine-tuned BLIP on the Flickr30k dataset to generate image alt-text, building a PyTorch training pipeline with beam search inference (~2.01 validation loss).
- Identified key accessibility limitations of automated captioning through controlled experiments, documented via walkthrough video and reproducible notebooks.

SKILLS

Programming: Python, C, C++, MATLAB, Java, SQL, HTML/CSS, React, JavaScript, TypeScript

ML, Data & Hardware: PyTorch, scikit-learn, Pandas, NumPy, scipy, Beautiful Soup, Arduino IDE, Sensor Calibration, SOLIDWORKS

Systems & Tools: Embedded Systems, POSIX Threads, TCP/IP Sockets, Multithreading, Excel, Azure, Intune, Git, NVivo, pytest